

ITAO5 - Activity.

- 1.) CCTV image contains Salt - and - pepper noise.
- * Technique: median Filter.
 - * Reason: It effectively removes Salt - and - pepper noise while preserving image edges.

- 2.) ~~Salt~~ Satellite image is corrupted by Gaussian noise.

- * Technique: Gaussian Filter
- * Reason: It reduces Gaussian noise by smoothing the image and improving overall quality.

- 3.) medical x-ray image has random noise, but fine edges must be preserved.

- * Technique: Bilateral Filter.
- * Reason: It removes random noise while preserving important edges and fine details in the x-ray image.

4). A Scanned document has black and white dots scattered randomly.

* Technique: median filter.

* Reason: It removes black and white (salt-and-pepper) noise while keeping text edges clear.

5). A low-light image shows heavy grain without clear boundaries.

* Techniques: Gaussian filter

* Reason: It reduces grain and produces a smoother image.

6). A foggy road image has very low contrast.

* Technique: Histogram Equalization

* Reason: It enhances image contrast, making objects and roads more visible.

7) An underwater image appears dark and washed out.

- * Technique: Contrast stretching
- * Reason: It improves brightness and color contrast, making the image clearer and more vibrant.

8) A chest x ray shows poor visibility of internal structures.

- * Technique: Histogram Equalization
- * Reason: It improves contrast and makes internal structures more visible.

9) A night-time traffic image lacks intensity variation.

- * Technique: Contrast stretching
- * Reason: It expands the intensity range, making dark and bright regions clearer.

10) A gray scale image uses only a narrow intensity range.

- * Technique: Contrast stretching
- * Reason: It increases the dynamic range and enhances the overall contrast.